GLAUCOMA

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Glaucoma = silent thief of sight! #2 cause of irreversible blindness!

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Glaucoma = optic neuropathy + field loss. IOP is a risk factor, NOT the definition!

DEFINITION

"A progressive optic neuropathy with characteristic disc & field changes; IOP is major risk factor." -- Khurana

Aqueous Humour Dynamics (VIVA FAVOURITE!)

Formed by:	Ciliary epithelium (active secretion 80% + ultrafiltration + diffusion)
Rate:	2-2.5 microL/min; total volume = 250 microL
Pathway:	Posterior chamber -> pupil -> anterior chamber -> drainage
Trabecular (90%):	TM -> Schlemm canal -> collector channels -> episcleral veins

Classification of Glaucoma

Type	Details
Primary OAG (POAG)	Most common glaucoma worldwide. Chronic, painless, bilateral
Primary ACG (PACG)	Acute or chronic angle closure. Common in India & Asia
Secondary OAG	Steroid-induced, pseudoexfoliation, pigmentary, phacolytic
Secondary ACG	Phacomorphic, iris bombe (post-uveitis), neovascular
Congenital	Buphthalmos (ox eye). Presents at birth or < 3 years
Normal Tension	Glaucomatous damage with IOP < 21 mmHg (vascular theory)

IOP Quick Facts

- Normal IOP = 10-21 mmHg (mean 16 +/- 2.5 mmHg). Diurnal variation < 5 mmHg
- IOP highest in early morning (4-6 AM) -- important for diagnosis!
- Ocular hypertension = IOP > 21 mmHg WITHOUT disc/field changes (monitor closely!)

POAG vs PACG table is asked in EVERY single exam paper!

POAG vs PACG -- The Comparison Table

Feature	POAG (Open Angle)	PACG (Angle Closure)
Onset	Insidious, chronic	Acute / subacute / chronic
Symptoms	Painless, late presentation	Severe pain, vomiting, halos
Angle	Open (Grade III-IV)	Narrow/Closed (Grade 0-I)
IOP	Mildly elevated (22-40)	Very high in acute (40-80 mmHg)
Pupil	Normal	Mid-dilated, fixed, oval
Cornea	Clear	Edematous (steamy cornea)
AC depth	Normal	Shallow
C:D ratio	Gradually increases	Rapid cupping if untreated
Conjunctiva	White, quiet eye	Circumcorneal congestion
Mechanism	Trabecular resistance	Pupillary block / plateau iris
Rx (1st line)	Prostaglandin analogs	Laser PI (peripheral iridotomy)

★ ACUTE ANGLE CLOSURE = OPHTHALMIC EMERGENCY! Do not miss!

Triad: Severe pain + Red eye + Hazy cornea. IOP 40-80 mmHg. Rx: IV Mannitol + Pilocarpine + Laser PI

Glaucomatous Optic Disc Changes (DRAW IN EXAM!)

- **Cup:Disc ratio** — > 0.6 is suspicious; > 0.8 = likely glaucoma. Asymmetry > 0.2 between eyes
- **Vertical elongation** — Cup elongates vertically FIRST (superior & inferior rim thin first)
- **ISNT Rule violated** — Normal: Inferior > Superior > Nasal > Temporal rim thickness
- **Disc hemorrhage** — Drance hemorrhage = splinter haemorrhage at disc margin (POAG sign)
- **Laminar dot sign** — Visible lamina cribrosa pores due to loss of nerve fibres
- **Bayoneting of vessels** — Vessels disappear at cup edge, reappear at base = deep cup

★ ISNT Rule Mnemonic

Inferior thickest, Superior next, Nasal, then Temporal thinnest. Violation = glaucoma suspect!

Investigations + Management

Tonometry + Gonioscopy + Perimetry = the holy trinity of glaucoma Dx!

Investigations for Glaucoma

Investigation	Details
Tonometry	GAT = gold standard. Normal IOP < 21 mmHg. Schiotz = indentation type
Gonioscopy	Shaffer grading (0-IV). Grade 0-I = occludable. Direct & indirect types
Perimetry	Humphrey (automated) = standard. 30-2 & 24-2. Arcuate scotoma = classic
Pachymetry	CCT: Thin cornea = higher risk. Thick cornea = falsely high IOP on GAT
Ophthalmoscopy	Cup:Disc ratio, ISNT rule, disc haemorrhage, NRR thinning
OCT / HRT	RNFL thickness measurement. Earliest objective damage detection
Provocative tests	Dark room test, prone test (for angle closure); water drinking test

Visual Field Defects in Glaucoma (Perimetry)

- **Bjerrum (arcuate) scotoma** — Most characteristic field defect. Arc from blind spot
- **Paracentral scotoma** — Small scotomas near fixation in Bjerrum area -- earliest
- **Seidel scotoma** — Extension of blind spot -- early arcuate scotoma
- **Ring scotoma** — Superior + inferior arcuate scotomas merge = double arcuate
- **Nasal step of Roenne** — Asymmetry at nasal horizontal meridian -- very specific
- **Temporal island** — End-stage: only temporal crescent of vision remains

Anti-Glaucoma Drugs (VIVA + MCQ GOLD!)

Drug Class	Examples	Mechanism	Key Side Effect
PG analogs	Latanoprost, Travoprost	Increase uveoscleral outflow	Iris colour change, lash growth
Beta-blockers	Timolol, Betaxolol	Decrease AH production	Bradycardia, bronchospasm
Alpha agonists	Brimonidine	Decrease prod + increase outflow	Allergy, drowsiness in kids
CAI (topical)	Dorzolamide, Brinzolamide	Decrease AH production	Metallic taste, stinging
CAI (systemic)	Acetazolamide	Decrease AH production	Metabolic acidosis, paraesthesia
Miotics	Pilocarpine	Open TM (contract ciliary m.)	Miosis, brow ache, myopic shift
Osmotic agents	IV Mannitol, oral Glycerol	Reduce vitreous volume	Acute angle closure emergency

Surgical Management

Surgery	Details
Trabeculectomy	Gold standard filtering surgery. Creates fistula: AC -> subconj space (bleb)
Laser PI	Nd:YAG peripheral iridotomy. DOC for acute & chronic angle closure
Laser trabeculoplasty	ALT/SLT for POAG. Opens trabecular meshwork. Repeat with SLT

Exam Strategy + High Yield

Glaucoma = guaranteed
15+ marks in ophthal!
You've GOT this Sammy!

Most Repeated LAQs (10-12 marks)

- 1. Define glaucoma. Classify. Describe POAG in detail (features + Rx)
- 2. Compare POAG vs PACG (table format with management)
- 3. Describe acute angle closure glaucoma -- clinical features & management
- 4. Enumerate anti-glaucoma drugs with mechanism of action (table)
- 5. Describe congenital glaucoma -- features, diagnosis & management

Most Repeated SAQs (5-6 marks)

- 1. Aqueous humour -- formation, circulation & drainage
- 2. Gonioscopy -- types, Shaffer grading, clinical significance
- 3. Visual field defects in glaucoma (with diagram)
- 4. Trabeculectomy -- indications, steps, complications
- 5. Provocative tests for glaucoma
- 6. Steroid-induced glaucoma / Pseudoexfoliation glaucoma

LAQ Answer Blueprint

1	Define	Optic neuropathy with disc + field changes (NOT just IOP)
2	Classify	OAG/ACG, primary/secondary, congenital
3	Compare	POAG vs PACG table (ALWAYS draw this!)
4	Investigate	Tonometry, gonioscopy, perimetry, OCT
5	Treat	Drugs table + surgical options
6	Indian	NPCBVI, blindness stats, glaucoma prevalence

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
#1 cause of irreversible blindness (world)	Glaucoma
Most common glaucoma worldwide	POAG (Primary Open Angle)
Most common glaucoma in India	PACG (higher in Asians)
Gold standard tonometry	Goldmann Applanation (GAT)
Most characteristic field defect	Bjerrum (arcuate) scotoma
DOC for POAG	Prostaglandin analogs (Latanoprost)
DOC for acute angle closure	IV Mannitol + Pilocarpine + Laser PI
Gold standard surgery for glaucoma	Trabeculectomy
Congenital glaucoma sign	Buphthalmos (ox eye), Haab striae
Steamy cornea + red eye + fixed pupil	Acute angle closure glaucoma
Glaucoma + normal IOP	Normal tension glaucoma (NTG)
Earliest visual field defect	Paracentral scotoma in Bjerrum area

High Yield Checklist